

Transport Layer: Introduction and Services

1 Introduction

The **Transport Layer** enables *logical communication* between **application processes** on different hosts. To applications, it appears as if processes are directly connected, regardless of physical distance or network complexity.

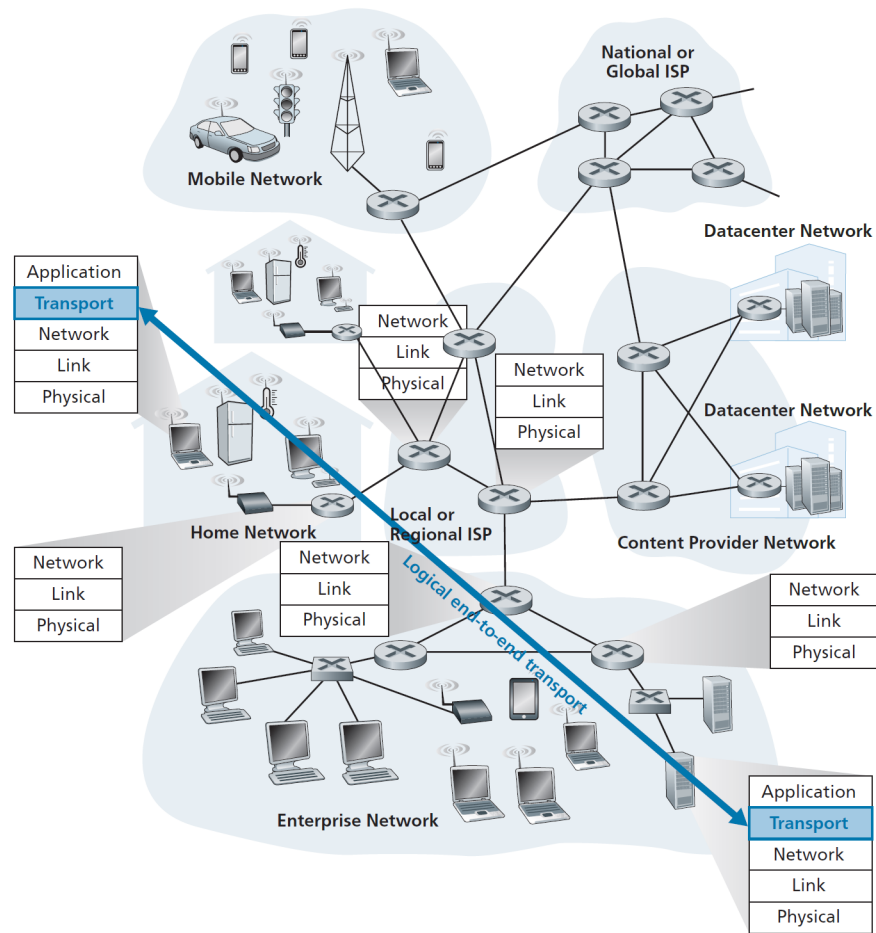
Note

The transport layer masks the details of routers, links, and infrastructure, providing seamless end-to-end communication.

On the **sender side**, messages are segmented, headers are added, and **transport segments** are passed to the network layer.

On the **receiver side**, these segments are reassembled and delivered to the correct process.

Routers operate only on **network-layer fields** and ignore transport-layer headers.



Logical end-to-end communication by the Transport Layer

2 Relationship Between Transport and Network Layers

- **Network Layer:** Logical communication between *hosts*.
- **Transport Layer:** Logical communication between *processes*.

Analogy: Postal Service and Families

Example

Consider two houses (hosts), each with many children (processes):

- The **postal service** delivers letters between houses $\Rightarrow$ Network layer.
- Ann and Bill distribute letters inside their house $\Rightarrow$ Transport layer.
- Cousins exchanging letters $\Rightarrow$ Application processes exchanging messages.

Just as Ann and Bill only manage mail inside their own house, transport protocols operate only within **end systems**, not within intermediate routers.

3 Service Models of the Transport Layer

Available Protocols

- **UDP (User Datagram Protocol)**
 - Connectionless and unreliable.
 - Provides only process-to-process delivery and basic error checking.
- **TCP (Transmission Control Protocol)**
 - Connection-oriented and reliable.
 - Guarantees in-order, correct delivery using sequence numbers, acknowledgments, and timers.
 - Provides **congestion control** to ensure fairness across the Internet.

Important

Key Difference: UDP is lightweight but unreliable. TCP is more resource-intensive but ensures reliability and fairness.

4 Constraints from the Network Layer

- Internet's network layer protocol: **IP (Internet Protocol)**.
- IP offers only a **best-effort service**:
 - No guarantee of delivery.
 - No guarantee of order.
 - No guarantee of data integrity.

- Therefore, IP is considered an **unreliable service**.

Implication: The transport layer's capabilities are constrained by IP. For instance, it cannot guarantee bandwidth or latency if IP does not.

Example

How Transport Layer Compensates:

- **Reliability:** TCP builds reliability on top of IP using acknowledgments and retransmissions.
- **Security:** Encryption (e.g., TLS) ensures confidentiality, even though IP cannot.

5 Summary of Transport Layer Services

- Provides logical communication between processes.
- Supports multiplexing and demultiplexing of messages.
- Ensures error detection.
- TCP extends services with reliability, flow control, and congestion control.
- UDP remains minimal, lightweight, and fast.